

Adaptive Conformal Inference through the Lens of Blackwell Approachability

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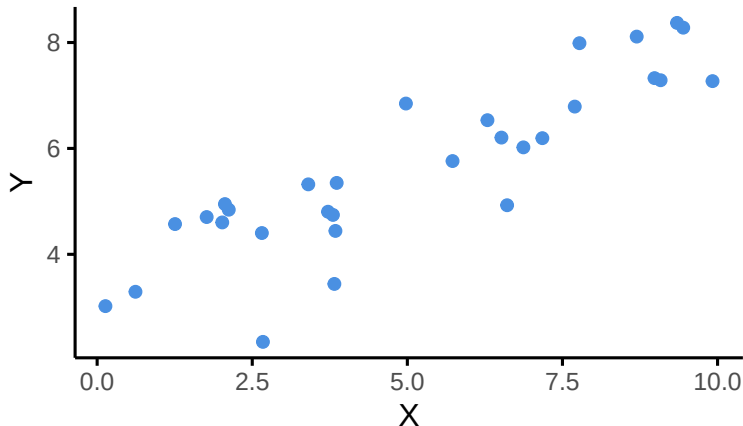


Outline

- 1. Introduction to Split Conformal Prediction [SCP]**
2. Introduction to Adaptive Conformal Inference [ACI]
3. ACI through the Lens of Blackwell Approachability
4. Blackwell Opportunistic ACI [BoACI]

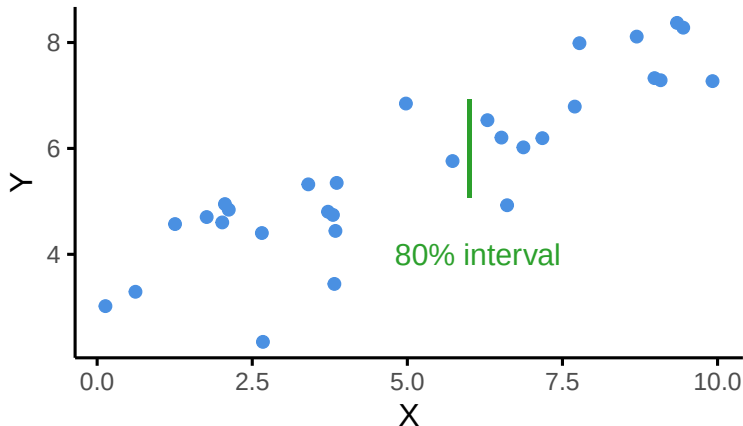
General setting for SCP

Regression setting with covariates $x_t \in \mathcal{X}$ and targets $y_t \in \mathbb{R}$. Past observations are available. We receive x_t and want to predict y_t with confidence $1 - \alpha = 80\%$.



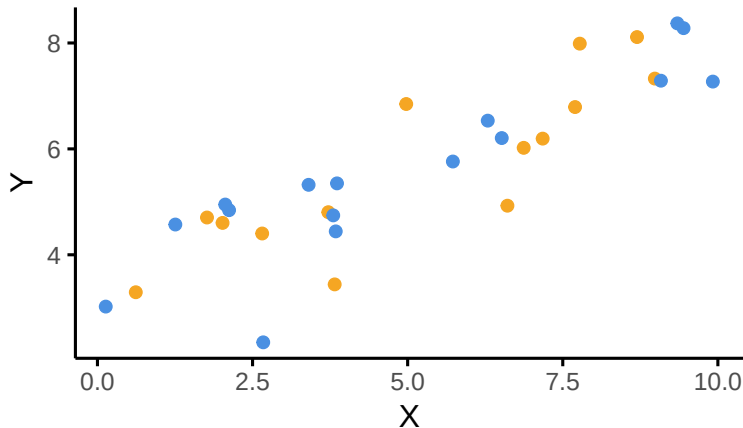
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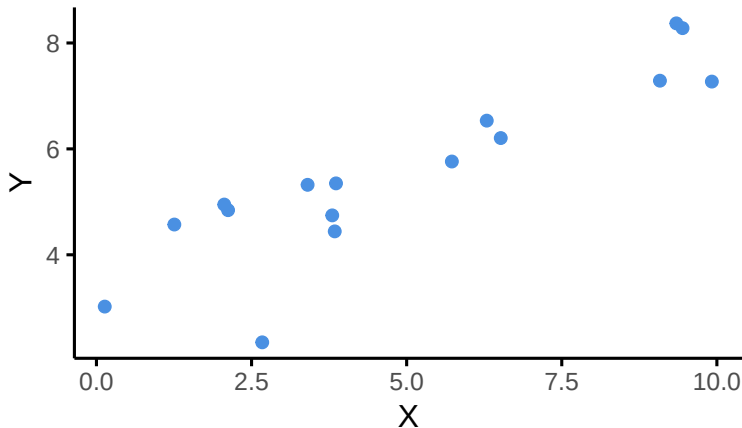
SCP with absolute residuals

Step 1. Split data
between $\mathcal{D}_{\text{train}}$ and $\mathcal{D}_{\text{calib}}$.



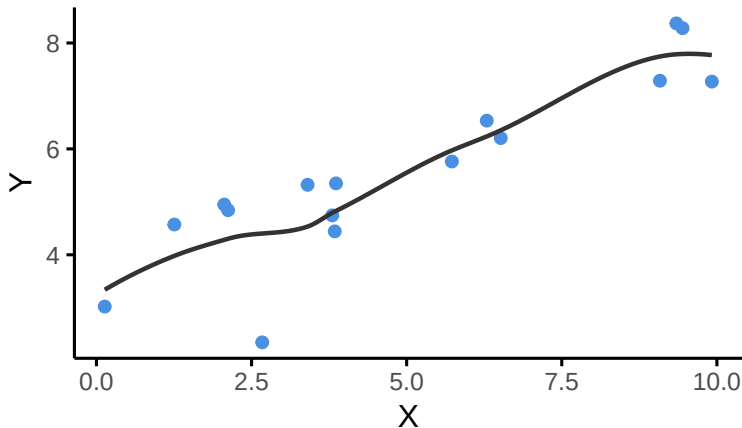
SCP with absolute residuals

Step 2. Learn a regression function $\hat{\mu}$ on $\mathcal{D}_{\text{train}}$.



SCP with absolute residuals

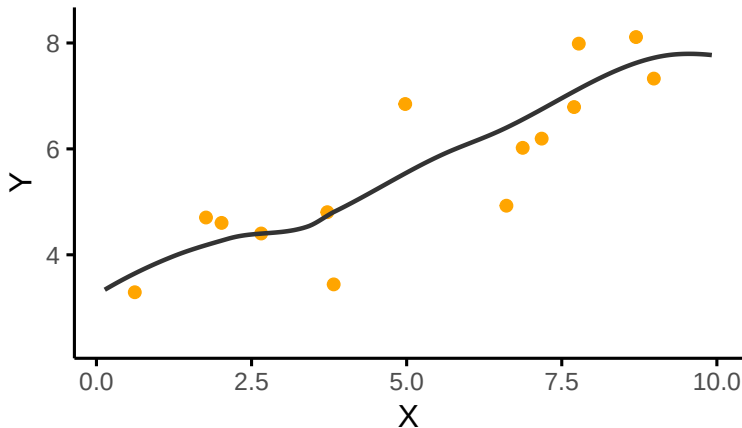
Step 2. Learn a regression function $\hat{\mu}$ on $\mathcal{D}_{\text{train}}$.



SCP with absolute residuals

Step 3. Compute non-conformity scores on $\mathcal{D}_{\text{calib}}$,

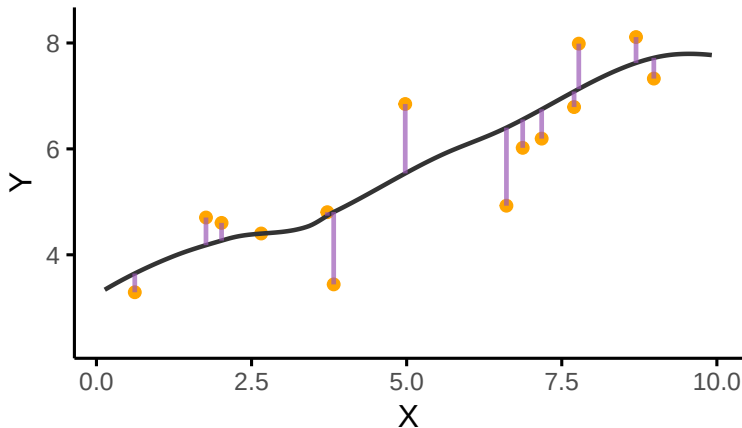
$$s_k = |y_k - \hat{\mu}(\mathbf{x}_k)|.$$



SCP with absolute residuals

Step 3. Compute non-conformity scores on $\mathcal{D}_{\text{calib}}$,

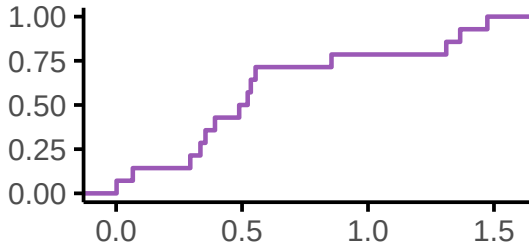
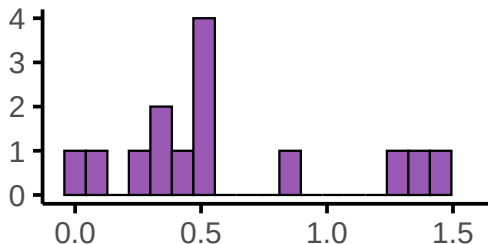
$$s_k = |y_k - \hat{\mu}(\mathbf{x}_k)|.$$



SCP with absolute residuals

Step 4. Compute empirical quantile q of the scores,

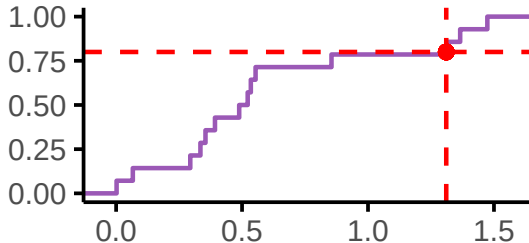
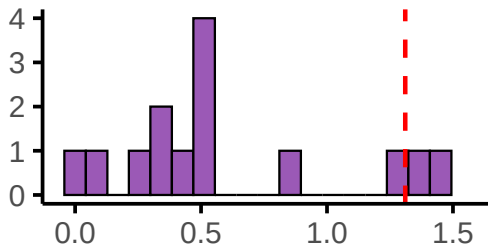
$$q = s_{\left(\lceil (|\mathcal{D}_{\text{calib}}|+1)(1-\alpha) \rceil\right)}.$$



SCP with absolute residuals

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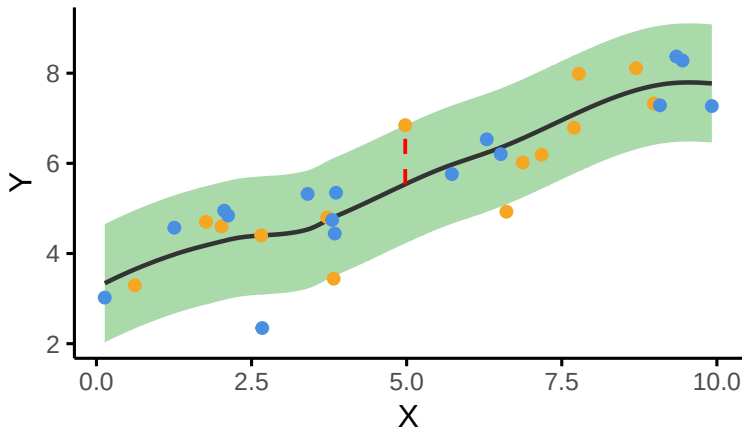
$$q = s_{\left(\lceil (|\mathcal{D}_{\text{calib}}|+1)(1-\alpha) \rceil\right)}.$$



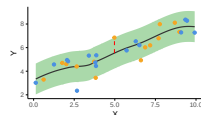
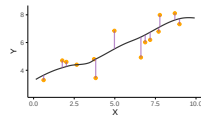
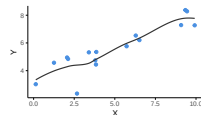
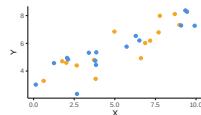
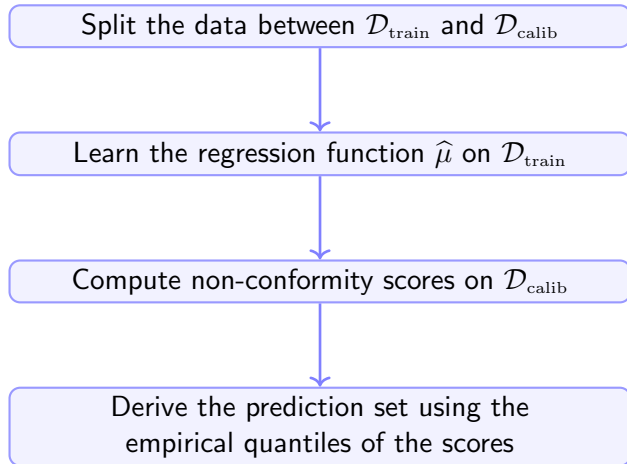
SCP with absolute residuals

Step 5. Output prediction set

$$C_{1-\alpha}(\mathbf{x}_t) = [\hat{\mu}(\mathbf{x}_t) \pm q]$$



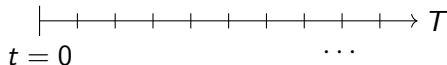
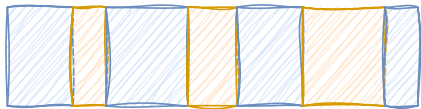
General summary of SCP



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New paradigm: sequential setting (Gibbs and Candès, 2021)



Split past data into $\mathcal{D}_{\text{train}}$ and $\mathcal{D}_{\text{calib}}$.

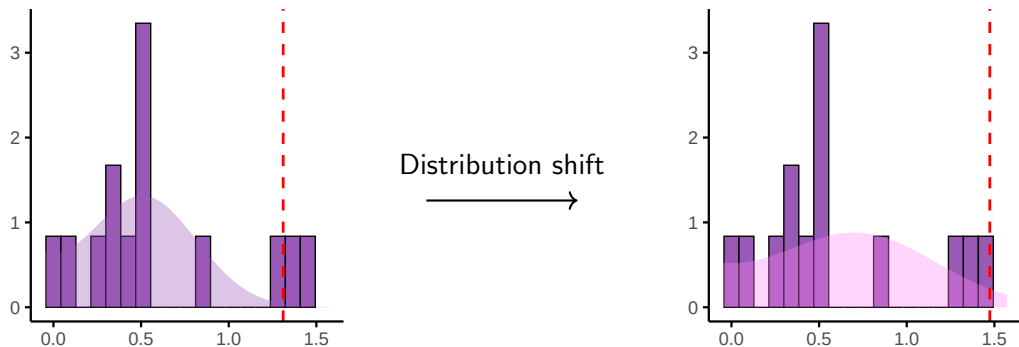
It induces candidate prediction sets $C(\cdot)$ based on $\hat{\mu}(\cdot)$ and on the empirical distribution of the non-conformity scores on $\mathcal{D}_{\text{calib}}$.

At each round $t \geq 1$, the learner

- observes x_t
- outputs a prediction set $C(x_t)$
- observes y_t and sees whether $y_t \in C(x_t)$ or not.

Beyond exchangeability: the need for adaptivity

Motivation: The i.i.d. assumption is not realistic in many applications, e.g., due to distribution shifts (such as during Covid).



The idea of Gibbs and Candès (2021) is to adapt the miscoverage level α at each round t by picking α_t based on past performance.

Clarification on the candidate prediction sets

Assumptions: We fix $\mathcal{D}_{\text{train}}$ and $\mathcal{D}_{\text{calib}}$ at the beginning of the game, and assume non-conformity scores are almost surely distinct and bounded by $L < \infty$.

The candidate prediction sets at round $t \geq 1$ are of the form:

$$C_{1-\alpha_t}(x_t) = \left[\hat{\mu}(x_t) \pm s_{\left(\lceil (|\mathcal{D}_{\text{calib}}|+1)(1-\alpha_t) \rceil\right)} \right], \quad \text{for some } \alpha_t \in [0, 1].$$

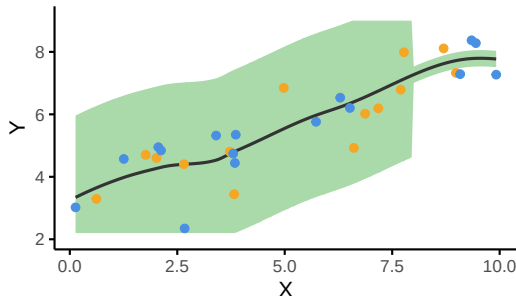
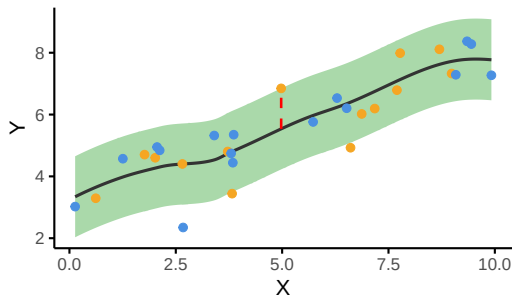
Therefore, in the setting we consider (Gibbs and Candès, 2021), our only degree of freedom is to pick α_t at each round t .

Remark on the edge cases

By convention, we set $s_{(0)} = 0$ and $s_{(|\mathcal{D}_{\text{calib}}|+1)} = L$, so that

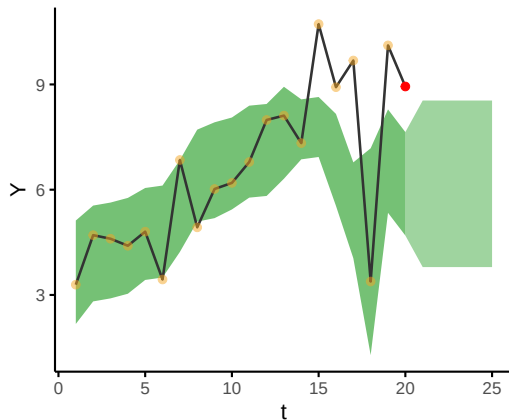
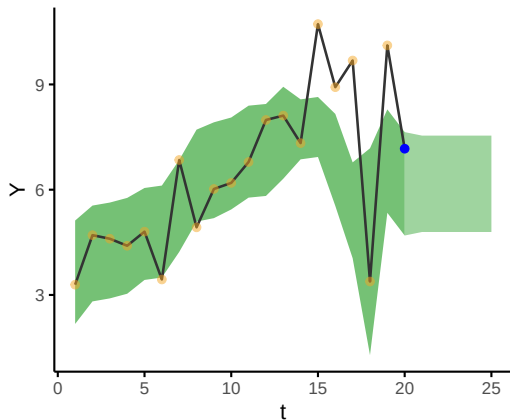
$$\begin{aligned}\alpha_t \geq 1 &\implies C_{1-\alpha_t}(x_t) = C_0(x_t) = \emptyset, \\ \alpha_t \leq 0 &\implies C_{1-\alpha_t}(x_t) = C_1(x_t) = [\hat{\mu}(x_t) \pm L].\end{aligned}$$

We can force the coverage frequency, but it may come at the cost of large prediction sets.



Presentation of the ACI algorithm (Gibbs and Candès, 2021)

$$\alpha_{t+1} = \alpha_t - \gamma(\mathbf{1}_{y_t \notin C_{1-\alpha_t}(x_t)} - \alpha), \quad \text{for some learning rate } \gamma > 0.$$



What guarantees for ACI?

The main theoretical result of Gibbs and Candès (2021) is that ACI achieves asymptotic validity even for arbitrary sequences $(x_t, y_t)_{t \geq 1}$, i.e., it satisfies

$$\limsup_{T \rightarrow \infty} \frac{1}{T} \sum_{t=1}^T \mathbf{1}_{y_t \notin C_{1-\alpha_t}(x_t)} \leq \alpha \quad \text{a.s.}$$

Extensions of ACI are mostly all about a sequential choice of the learning rate γ .

Open question:

Can we control the length of the prediction sets while ensuring validity?

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Definition of the objectives

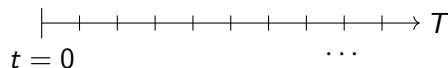
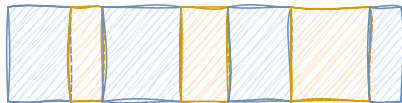
We aim to satisfy two objectives simultaneously:

$$\text{ensure} \quad \limsup_{T \rightarrow \infty} \frac{1}{T} \sum_{t=1}^T \mathbf{1}_{y_t \notin C_{1-\alpha_t}(x_t)} \leq \alpha, \quad (\text{Validity})$$

$$\text{control} \quad \limsup_{T \rightarrow \infty} \frac{1}{T} \sum_{t=1}^T \mathcal{L}(C_{1-\alpha_t}). \quad (\text{Efficiency})$$

To that end, we formulate ACI as a repeated **finite** game, and express these objectives in terms of an approachability problem.

Formulation of ACI as a repeated finite game



Split past data into $\mathcal{D}_{\text{train}}$ and $\mathcal{D}_{\text{calib}}$.

It induces a finite number of candidate prediction sets $C_{1-r(\alpha)}(\cdot)$ with $r(\alpha) \in \mathcal{A}$ based on $\hat{\mu}(\cdot)$ and on the empirical distribution of the non-conformity scores on $\mathcal{D}_{\text{calib}}$.

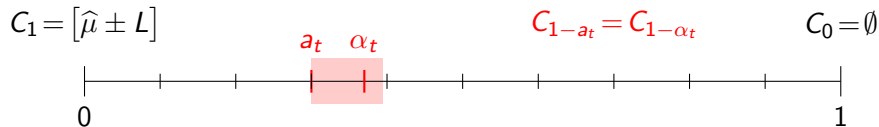
At each round $t \geq 1$, the learner

- observes x_t
- picks $a_t \in \mathcal{A}$ and outputs a prediction set $C_{1-a_t}(x_t)$
- observes y_t and a corresponding $b_t \in \mathcal{B}$ and sees whether $y_t \in C_{1-a_t}(x_t)$ or not based on a_t and b_t .

Formulation of ACI as a repeated finite game

Recall that a candidate prediction set is of the form

$$C_{1-\alpha_t}(x_t) = \left[\hat{\mu}(x_t) \pm s_{\left(\lceil (|\mathcal{D}_{\text{calib}}|+1)(1-\alpha_t) \rceil\right)} \right], \quad \text{for some } \alpha_t \in [0, 1],$$

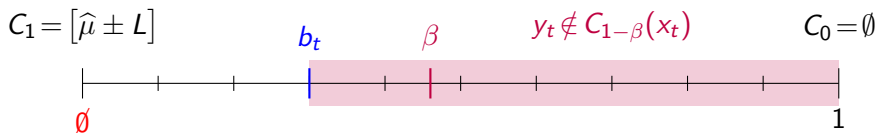


$$\mathcal{A} = \left\{ \frac{k}{|\mathcal{D}_{\text{calib}}|+1} : k = 0, 1, \dots, |\mathcal{D}_{\text{calib}}| + 1 \right\}$$

Formulation of ACI as a repeated finite game

Following Gibbs and Candès (2024), we can think about the target y_t as being generated by an opponent who picks some $b_t \in \mathcal{B}$ defined as

$$\begin{aligned} b_t &:= \sup \left\{ \beta \in [0, 1] : y_t \in C_{1-\beta}(x_t) \right\} \\ &= \min \left\{ \beta \in \mathcal{B} : y_t \notin C_{1-\beta}(x_t) \right\}. \end{aligned}$$

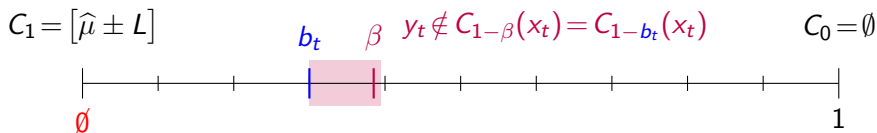


$$\mathcal{B} = \left\{ \frac{k}{|\mathcal{D}_{\text{calib}}|+1} : k = \emptyset, 1, \dots, |\mathcal{D}_{\text{calib}}| + 1 \right\}$$

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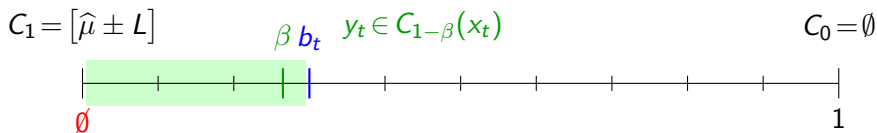


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$$\mathcal{B} = \left\{ \frac{k}{|\mathcal{D}_{\text{calib}}|+1} : k = \emptyset, 1, \dots, |\mathcal{D}_{\text{calib}}| + 1 \right\}$$

Formulation of ACI as a repeated finite game

This rewriting is useful to encode different settings:

Exchangeable setting.

At each round t , the opponent picks b_t uniformly at random in $\mathcal{B}$, i.e., plays

$$b_t \sim \mathbf{q}_{\text{unif}} = \frac{1}{|\mathcal{B}|} \sum_{b \in \mathcal{B}} \delta_b.$$

Adversarial setting.

At each round t , the opponent picks b_t according to an arbitrary distribution $\mathbf{q}_t \in \Delta(\mathcal{B})$.

In between settings.

?

Formulation of ACI as a repeated finite game

We recall our objectives (Validity) and (Efficiency)

$$\text{ensure} \quad \limsup_{T \rightarrow \infty} \frac{1}{T} \sum_{t=1}^T \mathbf{1}_{y_t \notin C_{1-\alpha_t}(x_t)} \leq \alpha, \quad (\text{Validity})$$

$$\text{control} \quad \limsup_{T \rightarrow \infty} \frac{1}{T} \sum_{t=1}^T \mathcal{L}(C_{1-\alpha_t}). \quad (\text{Efficiency})$$

By definition of b_t , they rewrite as

$$\text{ensure} \quad \limsup_{T \rightarrow \infty} \frac{1}{T} \sum_{t=1}^T \mathbf{1}_{b_t \leq a_t} \leq \alpha, \quad (\text{Validity})$$

$$\text{control} \quad \limsup_{T \rightarrow \infty} \frac{1}{T} \sum_{t=1}^T \mathcal{L}(C_{1-a_t}). \quad (\text{Efficiency})$$

Formulation of ACI as a repeated finite game

We can now rewrite (Validity) and (Efficiency) with the following vector-valued payoff function

$$m : (a, b) \in \mathcal{A} \times \mathcal{B} \mapsto \begin{pmatrix} \mathbf{1}_{b \leq a} \\ \mathfrak{L}(C_{1-a}) \end{pmatrix},$$

by studying the convergence of the average payoff

$$\overline{m}_T := \frac{1}{T} \sum_{t=1}^T m(a_t, b_t) \quad \text{to some target set } \mathcal{C} = \begin{pmatrix} [0, \alpha] \\ ? \end{pmatrix}.$$

This setting falls within the scope of Blackwell Approachability.

Which target set can be approached?

There exists a general strategy (i.e., that ignores the stochasticity of the opponent's play) that satisfies the following properties:

Generation mechanism	Minimal coverage	Maximal length
Exchangeable data	$1 - \alpha$	$\mathcal{L}(C_{1-\alpha})$
Adversarial data	$1 - \alpha$	—
ϵ -exchangeable data	$1 - \alpha$	$\mathcal{L}(C_{1-\alpha+\epsilon})$
Distribution shift	$1 - \alpha$	$\mathcal{L}(C_{1-\alpha'})$

Remark: The ϵ -exchangeable data setting includes AR(1) processes with weak correlation, and mild distribution shifts.

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What would we play if we knew the opponent's next play?

Assume (just for now) that we have access to $\mathbf{q}_t \in \Delta(\mathcal{B})$. What is the best action $a_t \in \mathcal{A}$ to play at round t ?

We define the function

$\text{mscv} : (\alpha, \mathbf{q}) \mapsto a \in \mathcal{A} \quad \text{s.t.} \quad C_{1-a} \quad \text{is the "smallest valid set, on average over } \mathbf{q} \text{"}.$

A natural choice is to pick $a_t = \text{mscv}(\alpha, \mathbf{q}_t)$.

Remark: This is what we do in the exchangeable setting since $\text{mscv}(\alpha, \mathbf{q}_{\text{unif}}) = r(\alpha)$.

Well, we don't have access to q_t ... but we can estimate it!

Idea: Use an auxiliary sequential probabilistic forecaster $\mathcal{F}$ of the opponent's play b_t to get a probabilistic forecast $\mathbf{z}_t \in \Delta(\mathcal{B})$ of b_t .

Precisely, we use a calibrated forecaster, i.e., that satisfies, for all Borel-measurable sets $G \subseteq \Delta(\mathcal{B})$,

$$\lim_{T \rightarrow \infty} \left\| \frac{1}{T} \sum_{t=1}^T \mathbf{1}_{\mathbf{z}_t \in G} (\mathbf{z}_t - \delta_{b_t}) \right\| = 0 \quad \text{a.s.}$$

Such forecasters exist, e.g., see Foster (1999); Mannor and Stoltz (2010) for a proof with Blackwell approachability theorem.

BoACI: a new (Blackwell opportunistic) strategy for ACI

The resulting procedure is based on opportunistic approachability (Bernstein et al., 2014).

BoACI strategy

Inputs: $\alpha \in [0, 1]$; auxiliary sequential forecaster $\mathcal{F}$ with outputs in $\Delta(\mathcal{B})$

Initialization: play any $a_1 \in \mathcal{A}$

At time steps $t = 2, 3, \dots$:

1. Get from $\mathcal{F}$ a probabilistic forecast $\mathbf{z}_t \in \Delta(\mathcal{B})$ of b_t
2. Play $a_t = \text{mscv}(\alpha, \mathbf{z}_t)$, observe x_t and output $C_{1-a_t}(x_t)$
3. Observe b_t (i.e., y_t), and feed $\mathcal{F}$ with b_t

Theoretical results for BoACI: Q -restricted opponent's play

Let $Q \subseteq \Delta(\mathcal{B})$ be any convex subset. The opponent is said to be Q -restricted if,

$$\lim_{T \rightarrow \infty} \frac{1}{T} \sum_{t=1}^T d(\mathbf{q}_t, Q) = 0 \quad \text{a.s.}$$

Examples.

- Exchangeable setting: $Q = \{\mathbf{q}_{\text{unif}}\}$ with $\mathbf{q}_{\text{unif}}$ the uniform distribution on $\mathcal{B}$.
- Adversarial setting: $Q = \Delta(\mathcal{B})$.
- ϵ -exchangeable setting: $Q = \{\mathbf{q} \in \Delta(\mathcal{B}) : d(\mathbf{q}, \mathbf{q}_{\text{unif}}) \leq \epsilon\}$ for some $\epsilon > 0$.
- Regime switching setting: $Q = \text{conv}\{\mathbf{q}_1, \dots, \mathbf{q}_K\}$ for some $\mathbf{q}_1, \dots, \mathbf{q}_K \in \Delta(\mathcal{B})$.

Theoretical results for BoACI: General statement

The following result is an informal statement, simplified for the sake of exposure.

Opportunistic Approachability of BoACI (Principato and Stoltz, 2025)

The BoACI strategy, when run with a calibrated forecaster, satisfies,

$$\bar{m}_T \longrightarrow \mathcal{C}_Q \subseteq \left(\begin{array}{c} [0, \alpha] \\ \max_{\mathbf{q} \in Q} \mathfrak{L}(C_{1-\text{mscv}(\alpha, \mathbf{q})}) \end{array} \right) \quad \text{as } T \rightarrow \infty, \quad \text{a.s.},$$

whenever the opponent is Q -restricted.

Comment on the result and insight on the tools

Comment: The results in the special cases (exchangeable, adversarial, etc.) are corollaries of the general result above, obtained by choosing Q accordingly.

Insight: The function m_{scv} is a response-function p^* that maps any $\mathbf{q} \in \Delta(\mathcal{B})$ to a **satisfactory** distribution over $\mathcal{A}$ (here, a pure action), i.e., such that

$$m^*(\mathbf{q}) = m(p^*(\mathbf{q}), \mathbf{q}) \quad \text{is a good payoff.}$$

The opportunistically approachable target set is defined¹ as

$$\mathcal{C}_Q = \text{conv}\left(\{m^*(\mathbf{q}) : d(\mathbf{q}, Q) = 0\}\right).$$

¹the formula is actually more complex for technical reasons, but this is the main idea.

Comparison of BoACI with the other ACI strategies

The following table summarizes the theoretical properties of the different ACI strategies, positioning BoACI within the literature.

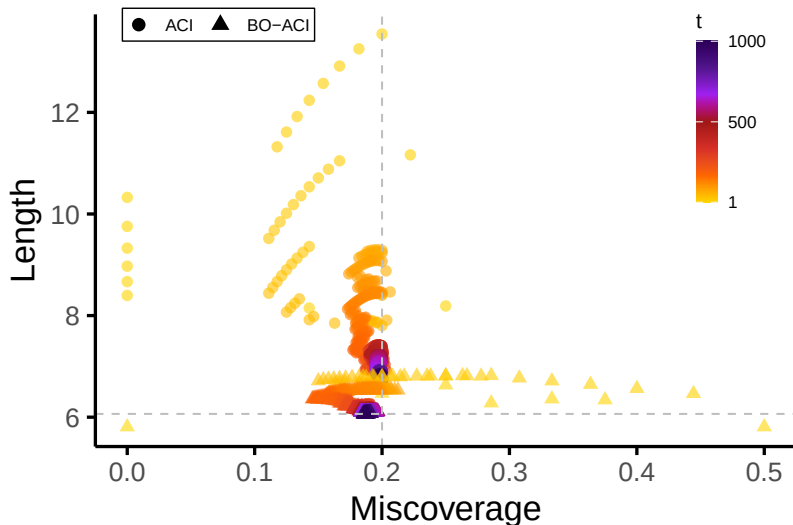
Theoretical properties	ACI ²	AgACI ³	DtACI ⁴	BoACI
Validity	✓	–	✓	✓
Efficiency (exchangeable case)	$\approx$ ³	–	–	✓
Efficiency (adversarial case)	–	–	–	–
Efficiency (intermediate cases)	–	–	–	✓
Computational complexity	✓	✓	✓	–

²Gibbs and Candès (2021), ³Zaffran et al. (2022), ⁴Gibbs and Candès (2024)

References I

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Comparison of BoACI with ACI in the exchangeable setting



Sketch of proof of Opportunistic Approachability

Step 1. The Q -restriction property propagates to the calibrated forecaster.

Step 2. Denote by $(\mathbf{z}_t)_{t \geq 1}$ the sequence of forecasts output by the calibrated forecaster.

$$d(\bar{m}_T, \mathcal{C}_Q) \leq \left\| \bar{m}_T - \frac{1}{T} \sum_{t=1}^T m^*(\mathbf{z}_t) \right\| + d\left(\frac{1}{T} \sum_{t=1}^T m^*(\mathbf{z}_t), \mathcal{C}_Q\right).$$

Step 3. We control (easily) the first term by calibration.

Step 4. We control the second term with Lemma 1 together with geometric arguments.